

MINI PROJECT REPORT
ON
“EVENT MANAGEMENT SYSTEM”

Bachelor of Technology
In
Computer Science and Engineering

Session 2024 - 2025

By

Shlok Rajvanshi (2300680100273)
Shorya Tyagi (2300680100274)
Shashank Mandoli (2300680100268)
Shishir (23006801002689)

Under the supervision of

Mr. Shivam Singhal

(Department of Computer Science and Engineering)



Meerut Institute of Engineering and Technology
NH -58, Baghpat Bypass Crossing, Meerut - 250005
(Affiliated to Dr. A. P. J. Abdul Kalam Technical University)

CERTIFICATE

This is to certify that mini project report entitled – “Event Management System” submitted by Shorya Tyagi, Shashank Mandoli, Shlok Rajvanshi and Shishir has been carried out under the guidance of Mr. Shivam Singhal, of Computer Science and Engineering, Meerut Institute of Engineering and Technology, Meerut. This project report is approved for Mini Project (BCC-351) in 3rd semester in department of Computer Science and Engineering, from Meerut Institute of Engineering and Technology, Meerut.

Supervisor :

Date :

DECLARATION

We hereby declare that the project entitled -“ Food Wastage Reduction Management System ”, which is being submitted as Mini project in department of Computer Science and engineering to Meerut Institute of Engineering and Technology, Meerut (U.P.) is an authentic record of our genuine work done under the guidance of Mr. Shivam Singhal, of Computer Science and Engineering, Meerut Institute of Engineering and Technology, Meerut.

Date :	Shorya Tyagi (2300680100274)
	Shashank Mandoli (2300680100268)
Place :	Shlok Rajvanshi (2300680100273)
	Shishir (2300680100269)

ACKNOWLEDGEMENT

I express my sincere indebtedness towards our guide Mr. Shivam Singhal, of Computer Science and Engineering, Meerut Institute of Engineering and Technology, Meerut, for his valuable suggestion, guidance and supervision throughout the work. Without his king patronage and guidance the project would not have taken shape. I would also like to express my gratitude and sincere regards for his kind approval of the project. Time to time counseling and advises. I would also like to thank to our HoD Dr. Vikas Srivastava, Department of Computer Science and engineering, Meerut Institute of Engineering and Technology, Meerut for his expert advice and counseling from time to time. I owe sincere thanks to all the faculty members in the department of Computer Science and engineering for their kind guidance and encouragement time to time.

Date :

Shorya Tyagi
Shashank Mandoli
Shlok Rajvanshi
Shishir

INTRODUCTION

The **Event Management System** is a web-based application that simplifies the planning, organization, and execution of events. Designed for flexibility and scalability, the system utilizes the **MVC (Model-View-Controller)** architecture to maintain a clean separation of concerns, making it easy to manage and extend functionalities. The platform is tailored to cater to event organizers, participants, and administrators, streamlining event-related operations such as creating, updating, and deleting events while maintaining a robust database backend.

OBJECTIVE

The Event Management System is designed to simplify the planning, organizing, and management of events. The system allows users to create, manage, and participate in events efficiently, reducing manual workload and increasing overall productivity. It caters to event organizers, participants, and sponsors, providing a seamless platform to handle all event-related operations.

The primary purpose of the Event Management System is to address the challenges faced in traditional event management practices, which often involve manual processes and inefficient communication channels. The system aims to:

- **Automate Event Operations:** Simplify tasks such as event creation, registration, updates, and cancellations.
- **Enhance Accessibility:** Provide a user-friendly interface for organizers and participants to interact with event details in real time.
- **Ensure Scalability:** Support events of varying sizes, from small gatherings to large-scale conferences.
- **Improve Efficiency:** Minimize administrative workload and enhance collaboration between stakeholders.

CHALLENGES ADDRESSED

The platform tackles the following challenges:

1. Eliminating paperwork and manual tracking.
2. Providing real-time updates and notifications.
3. Ensuring data consistency and security.
4. Delivering a scalable system to handle high traffic during event registrations.

FEASIBILITY STUDY

A feasibility study evaluates the potential success of the Event Management System by analyzing various aspects, including technical, economic, operational, and legal feasibility.

1. Technical Feasibility

- **Technology Stack:** Built using MongoDB, Express.js, React.js, and TypeScript for scalability, flexibility, and performance.
- **Architecture:** Follows the **MVC** model for a clean separation of concerns and maintainability.
- **Security & Scalability:** Incorporates user authentication and supports future scalability to handle high traffic.

Conclusion: The project is technically viable, leveraging modern and reliable technologies.

2. Economic Feasibility

- **Development Costs:** Utilizes open-source tools, reducing software licensing costs.
- **Operational Costs:** Scalable cloud hosting ensures cost efficiency.
- **Benefits:** Automation minimizes manual effort, reducing operational expenses.

Conclusion: The project is economically viable with a high return on investment.

3. Operational Feasibility

- **Ease of Use:** User-friendly interface and minimal training required.
- **Acceptance:** Meets user needs with efficient workflows and real-time updates.

Conclusion: The system is operationally feasible and user-centric.

4. Legal Feasibility

- **Data Privacy:** Complies with regulations like GDPR, ensuring user trust.
- **Policies:** Clear terms of use and privacy statements are established.

Conclusion: The project is legally feasible, adhering to required standards.

5. Schedule Feasibility

- **Timeline:** Development can be completed in 2-4 months, including testing.
- **Resources:** Sufficient expertise and resources are available.

Conclusion: The timeline is achievable and realistic.

Overall Feasibility

The **Event Management System** is technically, economically, operationally, legally, and schedule-feasible, ensuring a high likelihood of success.

METHODOLOGY OF WORK

The methodology for the Event Management System involves a structured approach, focusing on iterative development and modular implementation to ensure efficient progress and quality output.

1. Requirements Gathering and Analysis

- **Objective:** Understand the requirements of event organizers, participants, and administrators.
 - **Activities:**
 - Conduct user surveys and interviews.
 - Identify core functionalities (e.g., event creation, updates, participant management).
 - Define technical requirements, including system architecture, database design, and security.
 - **Deliverables:**
 - System Requirement Specification (SRS) document.
-

2. System Design

- **Objective:** Create a blueprint of the system to define its structure and workflow.
- **Activities:**
 - Develop the system architecture using the **MVC pattern**.
 - Create data flow diagrams (DFD) and entity-relationship diagrams (ERD) for database design.
 - Design wireframes and prototypes for the user interface.

- **Deliverables:**
 - System architecture diagrams.
 - Database schema and UI prototypes.
-

3. Technology Selection

- **Objective:** Choose the appropriate tools and technologies for development.
 - **Technologies:**
 - **Frontend:** React.js with TypeScript and JavaScript for responsive UI.
 - **Backend:** Node.js with Express.js for server-side logic.
 - **Database:** MongoDB for flexible, scalable data management.
 - **Other Tools:** Git for version control, and Postman for API testing.
-

4. Development

The development phase follows an **iterative approach**, breaking the work into smaller, manageable modules:

- **Module 1:** User authentication and role-based access control.
 - **Module 2:** Event creation, update, and deletion functionalities.
 - **Module 3:** Dynamic event listing with search and filter options.
 - **Module 4:** Integration of the notification and feedback system.
 - **Module 5:** Admin dashboard for monitoring and analytics.
 - **Activities:**
 - Code implementation.
 - Unit testing for each module.
 - **Deliverables:** Functional modules and system integrations.
-

5. Testing and Quality Assurance

- **Objective:** Ensure the system is free of bugs and performs as expected.
- **Activities:**
 - **Unit Testing:** Test individual components.
 - **Integration Testing:** Verify modules work together seamlessly.
 - **System Testing:** Evaluate the overall system performance.
 - **User Acceptance Testing (UAT):** Involve stakeholders to validate the system.
- **Deliverables:** Test reports and bug fixes.

6. Deployment and Launch

- **Objective:** Make the system accessible to users.
- **Activities:**
 - Host the application on a cloud platform (e.g., AWS, Heroku, or Vercel).
 - Perform final performance testing on the live environment.
 - Train end-users and provide a user manual.
- **Deliverables:**
 - Fully deployed Event Management System.

7. Maintenance and Support

- **Objective:** Ensure the system remains functional and up-to-date post-launch.
- **Activities:**
 - Monitor performance and resolve user-reported issues.
 - Roll out periodic updates for new features and improvements.
- **Deliverables:** Maintenance logs and updated versions.

Timeline for Planning of Work

Phase	Timeline
Requirements Gathering	Week 1
System Design	Week 2 - Week 3
Technology Selection	Week 4
Development (Module-Wise)	Week 5 - Week 9
Testing and Quality Assurance	Week 10 - Week 11
Deployment and Launch	Week 12
Maintenance and Support	Ongoing

WORKFLOW

The workflow of the system is divided into several key processes:

- **User Authentication and Role-Based Access Control:**
 - Users (organizers, participants, and administrators) log in or register.
 - Role-based access ensures that each user type accesses specific functionalities.
- **Event Management:**
 - Organizers create, update, and delete events.
 - Event details (e.g., date, time, venue, and description) are stored in the MongoDB database.
- **Event Browsing and Registration:**
 - Participants view available events, filter them by categories, and register.
 - Registered users receive confirmation and notifications.
- **Admin Oversight:**
 - Administrators monitor user activity, system performance, and feedback.
 - Admins generate reports for analytics and decision-making.

TECHNOLOGY BUCKET

The success of the Event Management System is heavily reliant on the modern technologies chosen for its development. Each technology plays a significant role in ensuring scalability, flexibility, and an optimal user experience.

Frontend Technology: React.js

Overview:

React.js is a popular JavaScript library for building interactive and dynamic user interfaces. It facilitates the creation of reusable components, ensuring a consistent look and feel across the application.

Key Features:

- Component-based architecture for modular development.
- Virtual DOM for fast and efficient UI updates.
- Integration with TypeScript for better code maintainability.
- Responsive design for a seamless experience across devices.

Use in the System:

- Design and development of user interfaces, including event listings, registration forms, and admin dashboards.
-

Backend Technology: Node.js with Express.js

Overview:

Node.js is a JavaScript runtime built on Chrome's V8 engine, designed for building scalable network applications. Express.js, a web application framework for Node.js, simplifies server-side development by providing a lightweight and flexible structure.

Key Features:

- Asynchronous and event-driven architecture for handling multiple requests efficiently.
- Middleware support for authentication, validation, and error handling.
- RESTful API design for seamless communication with the frontend.

Use in the System:

- Manages server-side logic, including API endpoints for CRUD operations.
 - Handles authentication and authorization using JWT.
-

Database: MongoDB

Overview:

MongoDB is a NoSQL database that provides a flexible schema for storing and retrieving data. It is well-suited for applications with complex or unstructured data requirements.

Key Features:

- Document-based storage for JSON-like data structures.
- Scalability to handle large volumes of events and users.
- High performance for real-time data access and updates.
- Built-in replication and failover mechanisms for data reliability.

Use in the System:

- Stores user profiles, event details, registrations, and feedback.
- Enables fast data retrieval for event browsing and management.

Programming Languages: JavaScript and TypeScript

Overview:

JavaScript is a versatile scripting language used for both frontend and backend development. TypeScript, a superset of JavaScript, adds static typing for better code quality.

Key Features:

- Dynamic and asynchronous programming capabilities with JavaScript.
- Type safety and autocompletion features with TypeScript.
- Cross-platform support for compatibility with different environments.

Use in the System:

- Implements frontend components, backend logic, and API integration.
- Enhances code reliability and reduces bugs with TypeScript.

Version Control: Git

Overview:

Git is a distributed version control system that tracks changes in code during development.

Key Features:

- Enables collaboration among developers through branching and merging.
- Tracks code history and facilitates rollbacks if needed.
- Integrates with GitHub for remote repository hosting.

Use in the System:

- Manages source code and enables team collaboration.
- Ensures version control for all development phases.

Deployment and Hosting Platforms

Cloud Platform:

- **AWS (Amazon Web Services), Heroku, or Vercel** are considered for hosting the application. These platforms provide scalability, reliability, and global availability.

Key Features:

- Easy deployment of the frontend and backend components.
- Load balancing to handle high traffic during event registrations.
- Automated CI/CD pipelines for faster updates and releases.

Use in the System:

- Hosts the React.js frontend and Node.js backend.
- Ensures uptime and smooth performance for users worldwide.

Other Tools

1. **Postman:** Used for testing API endpoints during development.
2. **JWT (JSON Web Tokens):** Ensures secure user authentication and session management.
3. **Tailwind CSS:** Simplifies responsive and modern UI design with utility-first styling.
4. **Bcrypt.js:** Encrypts user passwords for enhanced security.

TECHNOLOGY INTEGRATION

The Event Management System combines these technologies to deliver a cohesive and efficient solution. The architecture ensures:

- **Frontend and Backend Communication:** RESTful APIs enable seamless interaction between React.js and Node.js.
- **Database Connectivity:** MongoDB integrates with the backend for real-time data storage and retrieval.
- **Secure Authentication:** JWT secures user sessions, while bcrypt.js protects user credentials.

TECHNOLOGY USED

1. Hardware Resources

Development Machines:

- Minimum Configuration:
 - Processor: Intel Core i5 or equivalent
 - RAM: 8GB
 - Storage: 256GB SSD
- Recommended Configuration:
 - Processor: Intel Core i7 or equivalent
 - RAM: 16GB
 - Storage: 512GB SSD

Servers:

- Cloud-based hosting platform (e.g., AWS, Vercel, or Heroku).
-

2. Software Resources

Frontend Development:

- React.js: For creating a responsive user interface.
- TypeScript & JavaScript: For robust and scalable coding.
- Tailwind CSS: For styling and designing the UI components.

Backend Development:

- Node.js: For server-side scripting.
- Express.js: For API and server management.
- Bcrypt.js: For securing user passwords.

Database Management:

- MongoDB: For data storage and retrieval.

Version Control:

- Git: For source code management.
- GitHub: For collaboration and repository hosting.

Testing Tools:

- Postman: For API testing.
- Jest: For unit testing components.

Deployment Tools:

- Vercel or Heroku: For hosting the application.

REFERENCES USED

Web Development Technologies:

1. React.js Documentation: <https://reactjs.org/docs/getting-started.html>
2. Node.js Documentation: <https://nodejs.org/en/docs/>
3. MongoDB Documentation: <https://www.mongodb.com/docs/>
4. Express.js Guide: <https://expressjs.com/en/starter/guide.html>
5. Tailwind CSS Documentation: <https://tailwindcss.com/docs/>

Articles and Tutorials:

1. "Understanding the MVC Architecture": <https://www.freecodecamp.org/news/the-model-view-controller-pattern-mvc-architecture-and-frameworks-explained/>
2. "Building Scalable Applications with Node.js and MongoDB": <https://blog.logrocket.com/scalable-apps-with-node-js-mongodb/>

Tools and Platforms:

1. Git and GitHub Documentation: <https://git-scm.com/doc>
2. Vercel Deployment Guide: <https://vercel.com/docs>
3. Postman API Testing Documentation: <https://learning.postman.com/docs/getting-started/introduction/>

Other Online Resources:

1. Mozilla Developer Network (MDN) Web Docs: <https://developer.mozilla.org/>
2. W3Schools: JavaScript and HTML Tutorials: <https://www.w3schools.com/>
3. Stack Overflow Community: <https://stackoverflow.com/>

Research and Analytics:

Event Management Software Market Size & Trends: <https://www.marketresearchfuture.com/>